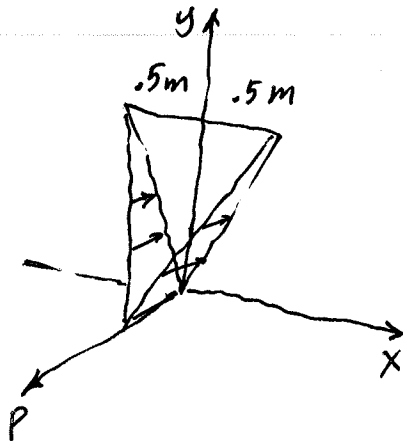
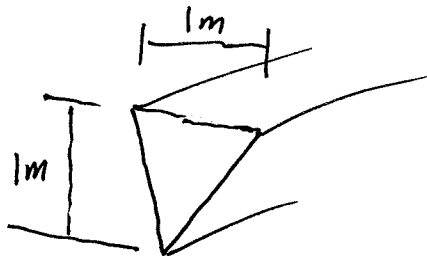
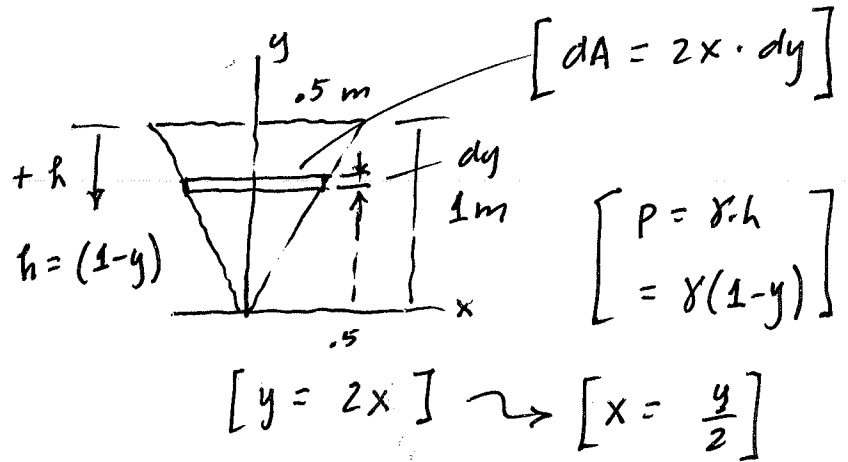


Ex 9.16

Integration



$$\rho_w = 1000 \text{ kg/m}^3$$



$$dF = dV = p \cdot dA$$

$$\text{Volume of } \nearrow = \gamma h dA$$

Pressure

$$\text{Distrib} = \gamma(1-y) 2\left(\frac{y}{2}\right) dy$$

$$dF = \gamma(y - y^2) dy$$

$$F = \gamma \int_0^1 (y - y^2) dy = 9810 \left[\frac{1}{2} y^2 - \frac{1}{3} y^3 \right]_0^1$$

$$= 9810 \left[\frac{3}{6} - \frac{2}{6} \right]$$

$$= \frac{9810}{6}$$

$$F = 1.635 \text{ kN}$$

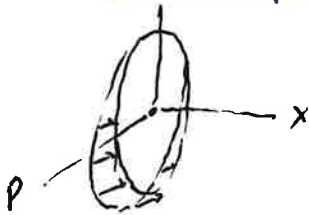
$$\text{To find } \bar{y} : \quad \text{Numerator} = \int_V \bar{y} dV = \gamma \int_0^1 y(y - y^2) dy$$

$$\bar{y} = y \quad = 9810 \left[\frac{1}{3} y^3 - \frac{1}{4} y^4 \right]_0^1 = 9810 \left[\frac{1}{12} \right]$$

$$\bar{y} = \frac{9810 (1/12)}{9810 (1/6)} = 6/12 = 0.5 \text{ m} = \bar{y}$$

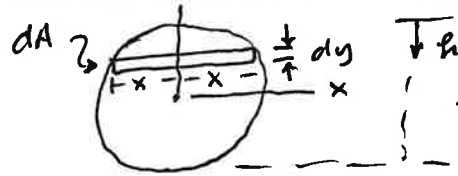
Integral on TI89: $\int ((.5-y) \times \sqrt{.25-y^2}), y, -.5, .5) = .19635$

1.5 Determine the magnitude of the resultant hydrostatic force acting on the flat end plate "C" of the tank. Density of water is 1.0 Mg/m^3 .



$$x^2 = .25 - y^2$$

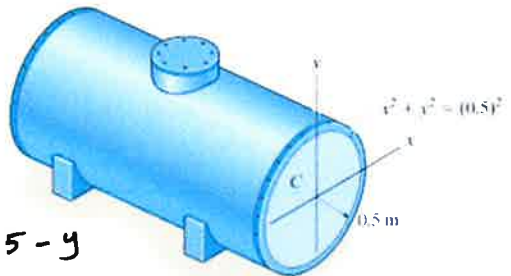
$$x = \sqrt{.25 - y^2}$$



$$dA = 2x \cdot dy$$

$$= 2\sqrt{.25 - y^2} dy$$

$$dp = p \cdot dA = 9810(.5 - y)(2)\sqrt{.25 - y^2} dy$$



$$F_R = \int_{-.5}^{.5} 9810(2)(.5 - y)\sqrt{.25 - y^2} dy$$

$$= 9810(2)(.19635)$$

$$F_R = 3.85 \text{ kN}$$

Section 2 – Moment of Inertia

2.1 Determine the moment of inertia for the shaded about the x axis.

$$I_x = \int_A \tilde{y}^2 dA$$

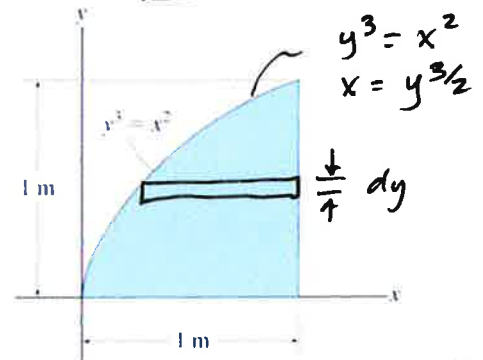
$$= \int_0^1 y^2(1 - y^{3/2}) dy$$

$$dA = (1 - x) dy$$

$$= (1 - y^{3/2}) dy$$

$$= \int_0^1 (y^2 - y^{7/2}) dy = \left(\frac{1}{3} y^3 - \frac{2}{9} y^{9/2} \right)_0^1 = \frac{1}{3} - \frac{2}{9} = \frac{1}{9} = 0.111 \text{ m}^4$$

$$= I_x$$



2.2 **TRUE** FALSE For a general area, the parallel axis theorem would be used when calculating I_y and using a horizontal differential element.

$\sim \perp$ to y axis.

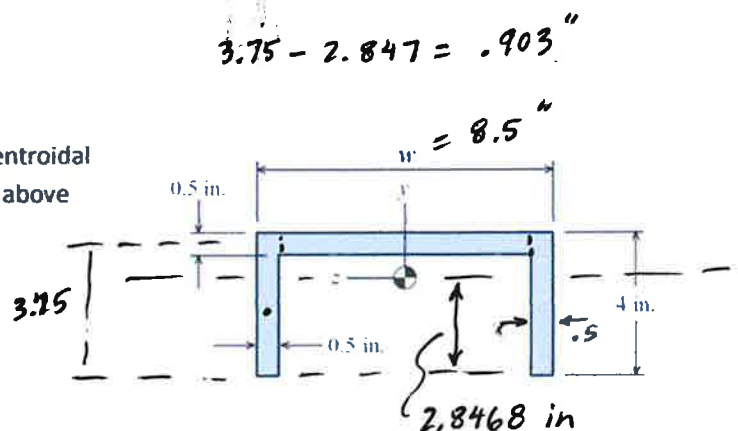
2.3 If $w = 8.50 \text{ in.}$, find the moment of inertia about the centroidal z-axis. The centroid of the shape is located 2.8468 inches above the bottom of the shape.

$$I_z = \sum (\bar{I} + Ad^2) =$$

$$= \left[\frac{1}{12} (7.5)(.5)^3 + 7.5(.5)(.903)^2 \right]$$

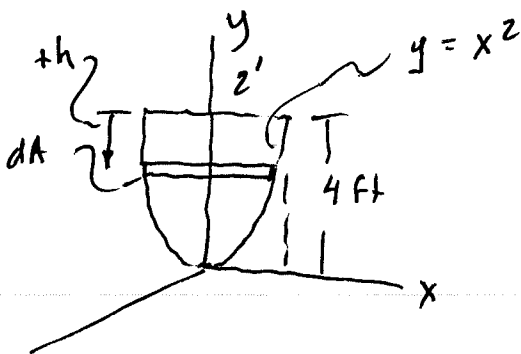
$$+ 2 \left[\frac{1}{12} (.5)(4)^3 + 4(.5)(.847)^2 \right]$$

$$I_z = 3.136 + 2(4.1015)$$



$$I_z = 11.34 \text{ in}^4$$

9-126 Integration



$$dA = (2x) dy$$

$$y = x^2 \Rightarrow x = y^{1/2}$$

$$dA = 2y^{1/2} dy$$

$$dF = dV = P \cdot dA$$

$$= \gamma(4-y) \cdot 2y^{1/2} dy$$

$$P = \gamma \cdot h$$

$$h = (4-y)$$

$$P = \gamma \cdot (4-y)$$

$$F = \int_0^4 \gamma (8y^{1/2} - 2y^{3/2}) dy$$

$$= 25 \left[\frac{2}{3} 8y^{3/2} - \frac{2}{5} 2y^{5/2} \right]_0^4$$

$$= 25 [42.67 - 25.6]$$

$$F = 426.7 \text{ lb}$$

Problem: What is γ ?

$$P = 100 = 4 \cdot \gamma$$

$$\gamma = 25 \text{ lb/ft}^3$$

Compare to water

$$\gamma = 62.4 \text{ lb/ft}^3!$$

Location, $\bar{y}$?

$$\tilde{y} = y$$

$$\text{Numerator: } = \int_V \tilde{y} dV = \gamma \int_0^4 y (8y^{1/2} - 2y^{3/2}) dy$$

$$= \gamma \int_0^4 [8y^{3/2} - 2y^{5/2}] dy$$

$$= 25 \left[\frac{2}{5} 8y^{5/2} - \frac{2}{7} 2y^{7/2} \right]_0^4$$

$$= 25 [102.4 - 73.14] = 731.5$$

$$\bar{y} = \frac{731.5}{426.7} = 1.714 \text{ ft} = \bar{y}$$